ADOBE TECHNICAL REPORT

Toward Physics-Faithful Video Generation

A depth-guided generator, a human-annotated flaw benchmark, and an auditable multi-tool critic

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REPORT STATUS AND SCOPE

Scope. This report describes three components built toward physics-faithful video generation: (1) a generator that renders a physics simulation into photorealistic video through a depth/structure control channel; (2) a human-annotated benchmark of prompt-versus-video flaws; and (3) an auditable multi-tool critic that detects such flaws. The three are designed to compose into a loop in which the critic's judgment refines the generator; **that closed loop is future work and is not evaluated here** (section 5).

What is and is not reported. We report the design of each component and the measured *descriptive* statistics of the artifacts (clip counts, flaw counts, generation configuration, model identifiers). We do *not* report critic accuracy, flaw coverage, false-alarm rates, a comparison between critic variants, or a learned video-quality score for the generator; those are performance claims deferred to a dedicated evaluation. Generator validation in this report is qualitative (visual inspection and adherence to the control signal), not a quantitative quality metric. Section 6 states the measurement protocol these results will follow.

Status vocabulary. BUILT present and exercised in the codebase; PARTIAL present with limited semantics; STUB present but inactive; NOT INTEGRATED built but not wired into the full loop; PROPOSED designed, not implemented.

Abstract

Text-to-video generators produce clips of high visual quality that nonetheless violate physical common sense—objects interpenetrate, motion ignores gravity, quantities fail to conserve—and often simply fail to depict what the prompt requested. Making generated video *physically faithful* requires three things at once: a generator that can be steered by physical structure rather than appearance alone; a way to measure, reliably and auditably, when a clip is wrong; and ground-truth data about the flaws real generators make. This report presents one component for each.

First, a **physics-guided generator**: a rigid-body simulation is rendered into a per-frame depth (or silhouette) control video and passed to the native control branch of a Wan 2.2-VACE video-diffusion backbone, so that scene geometry and motion are driven by the simulation while the text prompt supplies appearance. Second, a **human-annotated flaw benchmark**: 1,500 AI-generated clips carry human misalignment annotations—each with a severity, a time span, per-frame bounding boxes, the violated prompt fragment, and a free-text rationale—from which we derive a frozen, physics-focused evaluation set. Third, an **auditable multi-tool critic**: a served vision-language reasoner decomposes a prompt into checkable claims, routes each to a frozen open perception specialist that returns localized evidence only, and combines

the outcomes with deterministic rules into a three-valued verdict (*plausible, implausible, abstain*). The evaluation target is human labels, and no closed or proprietary model produces the critic's verdict.

The intended end state is a loop in which the critic supplies the reward that refines the generator. We describe each component and its honest implementation status, and we specify the human-label evaluation protocol under which the critic and the loop will be measured; comparative performance numbers are deferred to that evaluation.

1 Introduction

1.1 Physical faithfulness as the goal

State-of-the-art text-to-video models render strikingly realistic clips that still break physical common sense and often omit or distort what the prompt asked for. Physics-IQ quantifies the first problem directly: strong visual realism coexists with a large gap in physical understanding on its evaluated models and tasks [38]. Closing that gap is our long-term objective, and it decomposes into three coupled needs.

- **A steerable generator.** A model that follows *physical structure*—the geometry and motion of a scene—rather than reproducing the surface statistics of its training data.
- **A reliable, auditable detector of flaws.** A measurement instrument that says when a clip is wrong *and* shows why, at the level of individual evidence—because that judgment is the reward signal any improvement loop consumes.
- **Ground truth about real flaws.** A benchmark of the mistakes real generators make, annotated by people, against which any detector is measured.

1.2 Three contributions and the loop they form

This report contributes one component for each need and states how they are intended to compose.

1. **A physics-guided video generator** (section 2) that converts a rigid-body simulation into a depth or silhouette control video and drives a Wan 2.2-VACE diffusion backbone's native control branch, separating physical structure (from the simulation) from appearance (from the prompt).
2. **A human-annotated flaw benchmark** (section 3): a 1,500-clip corpus of AI-generated video with localized human misalignment labels, and a frozen physics-focused subset used as the evaluation target.
3. **An auditable multi-tool critic** (section 4) that decomposes a prompt into claims, routes each to an open perception specialist returning evidence only, and combines the evidence deterministically into a three-valued verdict.

The end state (section 5) is a closed loop: the benchmark trains and evaluates the critic, and the critic's verdict becomes the reward that refines the generator. *That loop is future work*; this report builds and characterizes its three parts.

Figure placeholder — generation pipeline.

rigid-body simulation → depth / silhouette control video → Wan-VACE control branch (control = depth, mask = all-white, prompt = appearance) → photorealistic clip.

Structure from the simulation; appearance from the prompt. To be rendered.

Figure 1: Physics-guided generation. The control video carries scene geometry and motion; the prompt carries appearance. **[TODO: replace with real frames: simulation, depth control, generated clip, side by side]**

1.3 Non-claims

We do not claim the closed generator–critic loop exists or that the critic has yet improved any generated video. We do not report critic accuracy, flaw coverage, false-alarm rate, or a comparison between critic variants; those await the evaluation of section 6. We do not report a learned video-quality score for the generator—its validation here is qualitative. And we claim novelty for none of the individual building blocks (structure-conditioned generation, prompt decomposition, specialist routing, three-valued abstention); the contribution is their integration toward physical faithfulness.

2 Physics-Guided Video Generation

2.1 Overview

The generator answers a specific question: can a video-diffusion model be made to follow the *geometry and motion* of a known physical scene, while a text prompt supplies only appearance? Our approach renders a physics simulation into a control video and feeds it to a diffusion backbone that accepts an explicit control channel, so structure comes from the simulation and texture from the prompt (fig. 1). The pipeline is: a rigid-body simulation → a per-frame depth (or silhouette) render → generation through the backbone's native control branch.

2.2 Backbone and control mechanism

The backbone is Wan 2.2-VACE-Fun-14B, a two-expert mixture-of-experts video-diffusion model with a native control branch, run through the `WanVACEPipeline` interface; because the two expert stacks exceed the memory of a single 80 GB accelerator, the experts are CPU-offloaded during generation. Generation is conditioned by passing the pre-computed control video into the pipeline's control input and an all-white mask into its mask input—the instruction to "generate the entire frame under the guidance of the control"—together with the text prompt for appearance and a single scalar *conditioning scale* governing how strongly the control branch binds. Crucially, the backbone consumes a *pre-computed* control video and does not re-estimate depth from an RGB source; the geometry is exactly the simulation's. This is a control-branch mechanism, not image-to-image denoising: it is neither an SDEdit-style noised re-synthesis [35] of a rendered source nor a copy

Setting	Value
Backbone	Wan 2.2-VACE-Fun-14B (two-expert MoE), diffusers `WanVACEPipeline`, bf16
Memory	experts CPU-offloaded (two-expert weights exceed an 80 GB accelerator)
Control signal	MuJoCo z-buffer depth (globally normalized); object silhouette for flat scenes
Conditioning	control video + all-white mask + text prompt; no reference image
Resolution	848×480 (flow-shift 3.0)
Frames / rate	81 frames at 16 fps
Sampler / steps	UniPC, 50 steps
Guidance scale	5.0
Conditioning scale	1.0 (swept over $\{0.5, 0.8, 1.0\}$)

Table 1: Deployed generation configuration, from the generation scripts. A small checkpoint provides a pre-flight sanity gate and is not used for final clips.

of source RGB, which is what distinguishes it from conditioning schemes that leak the source's synthetic appearance into the result.

2.3 Rendering the control signal

The control video is produced from a single simulation pass. We simulate rigid-body dynamics in MuJoCo [47] and, in the same pass, read the renderer's z-buffer for depth and its segmentation buffer for a silhouette. Depth is mapped to a near-bright grayscale under the inverse-depth convention the control branch expects, and—importantly—normalized *globally* across the whole clip from robust percentiles, not per frame, so that a receding object dims consistently instead of being re-stretched every frame. Most scenes use depth; flat tabletop scenes, where a single depth plane carries little signal, use the object silhouette instead. Structure-conditioned generation of this kind—driving synthesis from an explicit geometric control rather than from text alone—follows the lineage of ControlNet for images [57] and motion/structure control for video [25, 49]; here the control is a physically simulated depth field.

2.4 Pipeline and configuration

Table 1 lists the operating configuration. Generation runs as a staged GPU job: control videos are rendered, a small checkpoint provides a fast sanity gate, the 14B checkpoint generates the clips (data-parallel when more than one GPU is available), and the clips are streamed to storage.

2.5 What it achieves, and how we validated it

Generation was validated *qualitatively*: by visual inspection of the output clips and by a lightweight color-blob trajectory tracker that checks whether the generated object follows the control's path. There is no learned or quantitative video-quality score (no FVD, no human-preference model, no critic score) applied to this line, and we make no such claim. Under this qualitative bar, single-object scenes—a bouncing ball, a projectile arc—render cleanly with the object following the simulated trajectory. Multi-object interaction (e.g. a billiards break) is *not* solved: neither the deployed backbone nor the evaluated alternates produce a clean multi-object collision, and we report it as an open problem rather than a result.

2.6 Design choices

Two choices define the deployed recipe and are stated as design decisions. (1) **Native control over source re-synthesis.** A control branch that consumes a pre-computed depth video keeps the generator from importing a rendered source's synthetic look. (2) **No reference image.** The pipeline supports an optional appearance reference, but the deployed recipe omits it; including it degraded single-object trajectories in our inspection. A separate, in-progress study probes *when* during denoising the depth control can be released while the motion it has already induced is retained; its measurements are high-variance and not yet conclusive, and we do not report it as a settled result.

3 A Human-Annotated Video-Flaw Benchmark

3.1 The corpus

The benchmark is built from a corpus of **1,500 AI-generated video clips**, each carrying human annotations of where the video fails to match its prompt. The clips are AI-generated; the generating model is not recorded in the data, and we do not attribute them to a specific system. Every clip has at least one human-flagged flaw (table 2). This corpus is distinct from existing physical-commonsense benchmarks such as VideoPhy and its action-centric successor [5, 6], which we treat as separate transfer targets rather than as the annotation source here.

3.2 Human annotation schema

Each human flaw is a localized, typed record. The annotator marks, per flaw: a **severity** and **confidence**; a **time span** (start/end frame); **per-frame bounding boxes** localizing the flaw in space; the **prompt fragment** the video violates; and a free-text **rationale**. This is a richer object than a clip-level preference score: it says *what* is wrong, *where* in the frame, *when* in time, and *against which part of the prompt*. These fields are the human ground truth. A separate, coarse *scope* and *category* tag on each flaw is assigned automatically by a language-model ensemble, not by the human annotator, and we label it as machine-derived wherever it is used—it is convenience metadata, not part of the human ground truth. We record annotations as single-annotator; the corpus does not carry multiple independent annotations per flaw, and we make no inter-annotator-agreement claim.

3.3 The physics-focused evaluation set

The critic is evaluated not on all 1,500 clips but on a frozen, physics-focused subset, selected so that the flaws exercise physical and compositional reasoning rather than, say, subtitle typos. A physics-relevance filter keeps **606 clips with 1,107 flaws** as the full evaluation set, from which a

Set	Clips	Human flaws
Full corpus (all AI-generated, all annotated)	1,500	2,582
Physics-relevant pool	606	1,171
Full evaluation set (frozen)	606	1,107
Routine evaluation core (frozen)	150	306
continuity stratum within the core	29	85
Disjoint training pool (excludes the frozen sets)	711	—

Table 2: Benchmark statistics, from the dataset manifests. The full evaluation set drops a small number of audio-scoped flaws relative to the physics-relevant pool. The training pool excludes every clip in the frozen evaluation sets.

smaller **frozen core of 150 clips and 306 flaws** is drawn for routine measurement. The core is fixed by a recorded seed and date so that every experiment is scored against exactly the same target; a continuity-focused stratum (29 clips, 85 flaws) is tracked within it. Table 2 summarizes the counts. We are explicit that the evaluation set is the 606-clip / 1,107-flaw derivative (with a frozen 150-clip / 306-flaw core), not the full 1,500-clip corpus; conflating the two would overstate the evaluation. We are also explicit that the 150-clip / 306-flaw core is used as an *optimization set*: system and prompt decisions are repeatedly made against it, so coverage on it measures fit to that core rather than generalization, and a generalization claim requires a separately sequestered held-out set that has not influenced system design, which is future work.

3.4 Recall-oriented evaluation and a held-out training pool

Because every clip in the pool carries at least one flaw and there are no clean clips, the benchmark measures *recall*—did the critic find the human flaws—and cannot on its own measure precision or false-alarm rate; a "flag-everything" detector would score perfectly on recall. We state this limitation plainly and address false alarms in the protocol of section 6. For any learning that consumes this data, a training pool of 711 clips is held strictly disjoint from the frozen evaluation sets, so the evaluation core is never trained on.

4 The Auditable Multi-Tool Critic

4.1 Overview

The critic takes a prompt q and a video v and is designed to return a verdict in $\{\text{plausible}, \text{implausible}, \text{abstain}\}$ with a trace: the claims it checked, the evidence each tool produced, and the rule that combined them. A single holistic "is this plausible?" score would hide which observation drove the verdict and could confound photorealism with prompt satisfaction. Instead the critic decomposes: a served vision-language *reasoner* turns the prompt into small checkable claims and grounds each to the measurements it needs; frozen *perception specialists* return localized evidence only; and a *deterministic core* interprets most of that evidence with typed three-valued predicates and rolls the per-claim outcomes into the clip verdict. The learned parts are the reasoner and the specialists; the aggregation is deterministic and inspectable.

Capability	Open backend	Evidence returned
Object presence	LLMDet open-vocabulary detector [13]	presence; bounding box
Object count	CountGD-Box counter [2, 3] (SAM 2.1 internal)	cardinality of a named set
Kinematics	point tracking, TAPNext-style [59] + kinematic predicates	per-frame positions; static/-moving, direction, speed
Action timing	TimeLens-family temporal grounder [58]	time window of a named event
Spatial relation	unified SAM 3 masks [8] + monocular depth + reasoning	depth-ordered / image-plane relations
Holistic	*no tool by design* [20]	—
Audio†	FlexSED sound-event detection [18]	sound events with timestamps

Table 3: Perception specialists and the open backend wired to each. In the routed configuration the object, count, kinematics, action, and spatial-relation backends execute as stages; the object and count adapters declare stronger primaries that are not on the executed path, so the running backend is shown. The relation capability is served by a unified mask-plus-depth reasoning arm, and no holistic scoring tool is used in the verdict path by design. †The audio backend is wired in the typed-plan path and is not yet exercised in the routed configuration.

4.2 The reasoner

One served vision-language model—Qwen3-VL-30B-A3B-Instruct [39], BF16 weights (no quantization), tensor-parallel over two GPUs—performs the decomposition and grounding and runs three reasoner-mediated hard sub-checks (section 4.6). It is the only learned component that reasons over the whole prompt.

4.3 Perception specialists

Each claim is routed to a frozen open specialist that measures one thing and returns only that measurement (table 3). Two honesty notes are built into the table. First, some adapters *declare* a stronger primary backend that is not the one actually executed; we list the backend that runs. Second, there is no holistic scoring tool in the verdict path by design: a learned holistic video scorer is kept out so that the verdict is built from localized evidence and deterministic rules, though such a scorer could still serve as an external baseline.

4.4 Evidence-only tool contract

Specialists are frozen and constrained by a schema that forbids verdict-, label-, and mismatch-shaped fields: a specialist reports *what it measured* (a box, a count, a track, a time window, a depth), not *whether the prompt is satisfied*. A fail-closed adapter rejects any output that violates the schema, so a malformed or verdict-shaped output cannot enter the accepted evidence record, and each accepted measurement carries the prompt span and entity/event identifier that requested it. This guarantees that no specialist *directly emits* the verdict; it does not make a learned measurement semantically neutral, and a confidently wrong measurement can still mislead a downstream predicate.

4.5 Three-valued semantics and deterministic roll-up

Every claim resolves to supported, contradicted, or unknown, and unknown is first-class: when evidence is insufficient or ambiguous the critic abstains rather than guess. The gates are asymmetric—a positive, resolvable signal is required to assert contradicted, whereas a failed or empty detection becomes unknown, never proof of absence. The clip roll-up is a separate deterministic step: if any claim is contradicted, the clip is *implausible*; otherwise, if all claims are supported, it is *plausible*; otherwise the critic abstains. "Plausible" therefore means only that no checked claim was contradicted—a summary of the executed plan, not a certificate of complete physical verification.

4.6 Reasoner-mediated hard sub-checks

Three checks resist full symbolization and remain reasoner-mediated on tool-localized inputs: object subtype identity (is the detected object the specific kind named), dense action-event recall (did a named event occur, densely over time), and depth-ordered spatial relations. Each runs a fresh reasoner pass over evidence a tool has already localized, under the same asymmetric gate. We name them explicitly rather than imply a pipeline that is more symbolic than it is.

4.7 Auditable orchestration: a typed execution layer

Routing could be a flat claim-to-tool table. To make the orchestration inspectable, we instead compile each prompt into a typed *verification plan*—entities, event occurrences, observations (a tool operation plus its measurement contract), typed predicates, and claims—and type-check the plan before any tool runs. The validator rejects ill-typed plans loudly (for example, an existence check that reads an event clause rather than an entity noun, or a temporal check lacking two event localizations) and re-prompts the reasoner. Measurements that are described by an identical contract are executed once and shared; sharing uses an exact-key match only, so a whole-clip measurement and a sub-interval measurement are never conflated. Temporal relations are directional and tolerance-scaled: "A before B" requires a resolvable gap and returns unknown when the windows overlap within tolerance, following a subset of Allen's interval algebra [1].

Two execution configurations, stated plainly. The system exposes two configurations, and we keep them distinct. In the *routed* configuration, each claim is dispatched to its dedicated specialist (table 3) and the object, count, kinematics, action, and relation tools run as executable stages (the audio backend is wired but not yet exercised here). In the *typed-plan* configuration, the compiler and its typed predicates run, but the live value source used so far is the served reasoner over densely sampled frames, with cached and controlled values; wiring the typed-plan layer to dispatch the full specialist registry directly is integration work in progress, not an existing capability. Statements about running the named specialists refer to the routed configuration; statements about typed plans, validation, and reuse refer to the typed-plan layer. Global plan selection among alternative tool recipes and a value-of-information loop that spawns follow-up measurements are designed but **PROPOSED**, not built.

5 The Intended Refinement Loop

The three components are meant to close a loop: the benchmark trains and evaluates the critic; the critic scores generated clips; and the critic's verdict becomes a reward that fine-tunes or guides the generator toward physical faithfulness. **This loop is not yet built.** The generator and the critic today run on disjoint data—the generator on simulation-driven clips, the critic on the human-annotated benchmark—and no reward-driven generation run exists. Prerequisites are a critic whose reward is trustworthy on the frozen human labels (section 6) and a decision about where the reward enters generation (preference optimization, guidance, or data selection). We state the loop as the organizing goal and as explicit future work, not as a result.

6 Evaluation Protocol and Measurement

This section specifies how the critic will be measured; it deliberately reports no accuracy or coverage numbers.

Target and ground truth. The frozen physics-focused set of section 3 (150 clips / 306 flaws for routine measurement, 606 clips / 1,107 flaws for full sweeps). Human labels are the only ground truth. The 150-clip / 306-flaw core is the optimization set: because design decisions are repeatedly made against it, reported coverage measures fit to that core rather than generalization, and a generalization claim requires a separately sequestered held-out set that has not influenced system design.

Matching is outside the critic. To decide whether a critic output corresponds to a human-flagged flaw, a separate protocol uses two judge models (a GPT-family and a Gemini-family model, majority over runs). These judges perform *flaw matching only*; they are not components of the critic, do not produce its verdict, and are not treated as ground truth. Any "majority-of-three" in this project refers to such a model-vote protocol, never to human annotators.

Recall, and the false-alarm gap. Because the benchmark has no clean clips, the primary measurable quantity is recall of the human flaws. A meaningful false-alarm or precision number requires either clean clips or per-allegation adjudication of the critic's non-matching outputs; we specify that adjudication as part of the protocol and do not substitute a recall number for it.

Dense-frame requirement. The critic must observe the video densely. The evaluated components extract frames densely under a logged sampling policy, and any measurement obtained from a sparsely sampled input is excluded. A re-run counts only if it uses the dense path on the unchanged frozen target with BF16 tools at pinned revisions.

The pre-registered comparison. We evaluate the orchestration design with a paired, four-cell nested ablation on the 150-clip optimization set. The cells successively separate three possible sources of an observed difference: (i) the *wording of the verification question*—whether a claim is true of the video, versus whether the described content ever appears; (ii) the *decomposition* of the prompt into typed claims; and (iii) the *execution structure* itself—shared measurements, routing to capabilities, composition across measurements, and conflict resolution. The first two structured cells keep independent per-claim execution, so the final comparison isolates the contribution of the plan-based execution rather than crediting question-wording or decomposition gains to it.

Because tools differ substantially in cost—a detector call, a tracker call, and a full vision-language judgment are not one unit each—we record model and tool calls, processed frames, and GPU-seconds rather than treating every invocation as equivalent. Systems are compared at their natural budgets and through reciprocal budget curves: the plan-based system is capped at the per-claim baseline's budget, and the per-claim baseline is extended to the plan-based system's natural budget, under a priority policy fixed in advance. Allegations not matched to a human flaw count as *review burden* in the frozen-benchmark scoring, not as false positives; any separate adjudication of them is reported only as a benchmark-incompleteness diagnostic and does not expand the frozen labels or change the frozen-benchmark score. Uncertainty is estimated by resampling whole clips, because flaws within a clip are dependent.

No comparative results are reported here. The final report will disclose that this set was inspected and used during optimization, that the question-wording-controlled baseline was introduced after treatment outputs had been observed, and that no retroactive held-out split exists within these 150 clips; a claim of generalization therefore requires evaluation on a new frozen set.

7 Related Work

7.1 Physics and structure in video generation

Physical-commonsense generation and its evaluation are studied by VideoPhy [5, 6], PhyGenBench [36], and Physics-IQ [38]; V-JEPA-style work argues physics is best judged in a learned state space [14]. Our generator does not learn physics: it *imports* it from a simulator and asks the backbone to render it, via structure conditioning. Structure- and motion-conditioned synthesis has a strong lineage—ControlNet for images [57], motion control for video [49], the Wan family and its VACE control interface [25, 48], and, as guided editing rather than control-branch conditioning, SDEdit [35]. DiffPhy conditions a diffusion backbone on a physics chain-of-thought [56]; we instead condition on a rendered depth field from a simulation.

7.2 Evaluating text-to-visual faithfulness

Decomposing a prompt into checkable pieces is established: TIFA [22] and the Davidsonian Scene Graph [9] score generated questions/propositions with a vision-language model, VQAScore uses an image-to-text model [31], and GenAI-Bench, VBench, T2V-CompBench, FETV, and EvalCrafter benchmark compositional alignment [23, 28, 32, 33, 45]; embedding and preference scorers predict alignment or human preference [21, 27, 53], and learned quality models predict human judgment directly [19, 20]. We keep such a holistic learned scorer out of the verdict path. Relative to these, our critic routes each claim to a dedicated open perception specialist under an evidence-only contract, validates a typed plan before execution, distinguishes event occurrences, reuses perception only under an exact contract, and preserves claim-level provenance with an explicit unknown outcome. We claim novelty for the integration, not the parts.

7.3 Visual programs, tool-use, and modular video QA

Composing perception modules for reasoning includes Neural Module Networks [4], VisProg [17], ViperGPT [46], CodeVQA [44], and neural-symbolic VQA [24, 55]; LLM tool-controllers and plan-execute graphs include HuggingGPT, TaskMatrix.AI, ControlLLM, LLMCompiler, and ReWOO [26, 30, 34, 43, 52], with search-based variants [7, 54, 60, 61]; modular video QA includes ProViQ, MoReVQA, the VideoAgent systems, and VideoTree [10, 12, 37, 50, 51]. Our execution layer is narrower and verification-specific: a *typed* plan, type-checked before tools run, whose evidence obligations carry three-valued semantics; it does not search over alternative plans. Reuse of equivalent measurements is common-subexpression elimination in the multi-query-optimization sense [15, 16, 41, 42], and the specialists themselves draw on open detection, counting, tracking, grounding, and audio models [2, 3, 8, 11, 13, 18, 29, 40, 58, 59].

7.4 Flaw and preference datasets

Prior video benchmarks largely provide clip-level physical-commonsense or preference labels [5, 6, 36]. Our benchmark instead provides *localized* human flaw annotations—severity, time span, per-frame boxes, the violated prompt fragment, and a rationale—so that a detector can be scored not only on whether a clip is flawed but on whether it found the specific, spatially and temporally located error a person marked.

8 Limitations

Generation. Validation is qualitative (visual inspection plus a trajectory tracker); there is no learned quality metric for this line. Multi-object interaction is unsolved. The control signal comes from a simulator, so scene diversity is bounded by what is simulated, and the mapping from simulation to prompt appearance is not itself verified for physical consistency.

Benchmark. The corpus is recall-only (no clean clips), so precision is not directly measurable; flaws are single-annotator with no inter-annotator agreement; the scope/category tags are machine-derived; and the source generator of the clips is unrecorded. Repeatedly designing against one frozen set risks overfitting to it even without training on it.

Critic. The spatial relation predicate and negative-event verification are limited; the typed-plan layer does not yet dispatch the full specialist registry; binding remains a central failure mode, since the compiler enforces symbolic identifier consistency but cannot guarantee that heterogeneous tools localized the same physical instance; and the reasoner-mediated sub-checks reduce but do not remove semantic opacity.

Loop. The generator–critic loop is unbuilt, and critic quality on the frozen labels is the gate before any reward-driven generation is attempted.

9 Conclusion

Physical faithfulness in generated video requires steering, measurement, and ground truth together. This report contributes one piece for each: a generator that imports geometry and motion from a simulation and renders them through a video-diffusion control branch; a human-annotated benchmark that localizes real flaws in space, time, and prompt reference; and an auditable multi-tool critic that decomposes a prompt, routes each claim to an open specialist returning evidence only, and combines the evidence deterministically into a three-valued verdict. We have described each component and its honest implementation status and specified the human-label protocol under which the critic—and eventually the loop that lets it refine the generator—will be measured. The comparative numbers, and the closed loop itself, are the next milestones, not claims of this report.

A Appendix

To be expanded: the full generation configuration and control-render details; the benchmark schema and per-stratum statistics; the critic's plan schema, predicate type rules, and temporal-relation truth tables; the specialist license and revision manifest; and the pre-registered evaluation manifest.

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